What the 15 studies show

One-page summary | 13 October 2026

Mechanistic interpretability studies how a model produces its answers. These fifteen studies changed internal parts and measured the results. Their main lesson is clear: an output change alone does not show how the model solves a task.

Main findings

- **Check task ability first.** An internal part can affect the output even when the model rarely gives the correct answer. In one copying test, the requested answer ranked first in only 1 of 1,024 cases. A positive measured effect remained.
- **Define the exact change.** Different ways to remove or replace the same internal part can produce different results. Tests must match the proposed explanation.
- **Measure the claimed behavior.** A larger output change does not mean a better answer. A group of features can carry useful information that one selected feature does not. An effect that lasts does not establish useful memory.
- **Report mixed results.** Several proposed improvements gave no clear or consistent benefit. Giving rare cases more training weight was worse in five of six main feature comparisons.
- **Separate uncertainty from error.** In one numerical study, initial bounds could not establish 18.66% of nonzero estimated effect directions, averaged over models. Further calculation proved five initial directions wrong. The first percentage was not an error rate.

What is new

The contributions are specific controlled tests and complete records: code, results, saved models or model versions, and calculation checks. Examples separate coordinate changes from editing rules, compare two meanings of spike deletion, and check numerical effect directions.

The review connects these results and gives an ordered research plan. It adds no experiment. The studies do not establish fifteen new general principles.

Limits

The models are small, and many tasks are artificial. Five studies share one language model; three share a digit dataset. The results are not fifteen independent confirmations and do not measure failure rates in large deployed models.

Some models failed task requirements. Scores have different meanings and cannot be pooled. Rechecking saved models does not repeat training. The work is AI-authored, with separate AI checks. No human peer review is recorded.

Next research steps, in order

1. **Establish task ability.** Set success rules before testing. Keep failed runs.
2. **Test intended and unwanted effects.** For a specific-variable claim, require the intended change and preserve unrelated behavior.
3. **Test new conditions.** Fix the explanation before testing new models, words, and tasks.
4. **Test later behavior.** Separate fixed continuations from text generated by the model.
5. **Measure value and cost.** Report absolute errors, numerical uncertainty, and total computation.
6. **Seek external replication and human review.** Repeat training as well as calculations.

Source: Full review and study references.